

The Longevity Stack

Creatine · cold exposure · fasting · aerobic training · resistance training · melatonin

Mechanisms, human evidence, and hypotheses about what they do together

Research brief · 12 October 2026

Evidence is graded; speculation is labeled as speculation.

At a glance

Six interventions, one question: what does each actually do, how good is the evidence in humans, and what might they do *together*? The short version is that the stack is not six equal pillars. Two of the six carry almost all of the hard-outcome evidence; the other four are better understood as modulators, each with a specific job and a specific way to go wrong.

Five things worth knowing

1. **Aerobic fitness and resistance training are the foundation.** Cardiorespiratory fitness has the steepest, most consistent mortality gradient in preventive medicine with no observed ceiling[1,2]; any resistance training is associated with roughly 15% lower all-cause mortality, with most of the benefit at only 30–60 minutes a week[3,4].
2. **Creatine is the best-evidenced supplement here, mostly for muscle.** The brain effects are real at the level of phosphocreatine, but the cognitive benefits in healthy adults are modest and mixed; the clearest signal is under energetic stress such as sleep loss[5--9].
3. **Cold and fasting are mostly metabolic and subjective stories.** Cold produces a large, brief noradrenaline surge and small metabolic adaptations[10,13,14]. Time-restricted eating matches ordinary calorie restriction when calories are equated[19--21], with a mechanistic case for more.
4. **Melatonin is a clock signal, not a vitamin.** It has a clear job (timing), a plausible secondary job (mitochondrial antioxidant at higher doses), and an unresolved long-term safety signal in observational data[22].
5. **The interactions matter as much as the effects.** Post-lifting cold blunts hypertrophy[12]; antioxidant-dose melatonin near training may blunt training signals; and intensity, not fasting, drives the BDNF response[18]. Sequencing is a bigger lever than adding a seventh intervention.

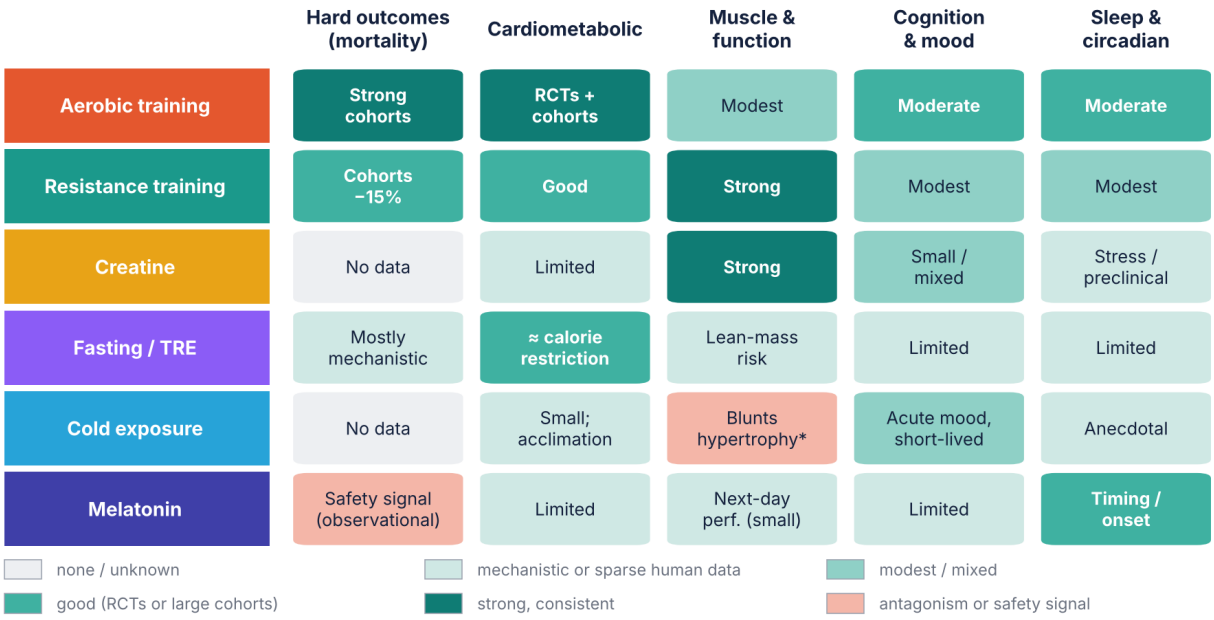


Figure 1. Evidence scorecard by outcome domain. Scores are my qualitative judgment of *human* evidence quality from the sources reviewed here, not a formal grading. *Cold blunts hypertrophy specifically when applied after lifting.

How to read this report. Chapters 2--7 summarize each intervention and cite sources in square brackets (numbered in the reference list at the end; items marked † could not be checked against the primary text). Chapter 9 is deliberately speculative; each hypothesis carries two ratings: *mechanistic plausibility* and *human evidence for the combination*. The second is almost always low, because almost nobody has tested these combinations.

1 The shared mechanism map

The interventions look unrelated, but they converge on a small number of cellular and neural switches. Energy stress activates AMPK and, downstream, PGC-1α, which drives mitochondrial biogenesis. Mechanical loading activates mTORC1 for muscle protein synthesis. Exercise and fasting both produce metabolites (lactate, β-hydroxybutyrate) that signal to the brain. Cold recruits the noradrenergic system, and melatonin sets the timing of the whole sequence. Figure 2 traces these paths.

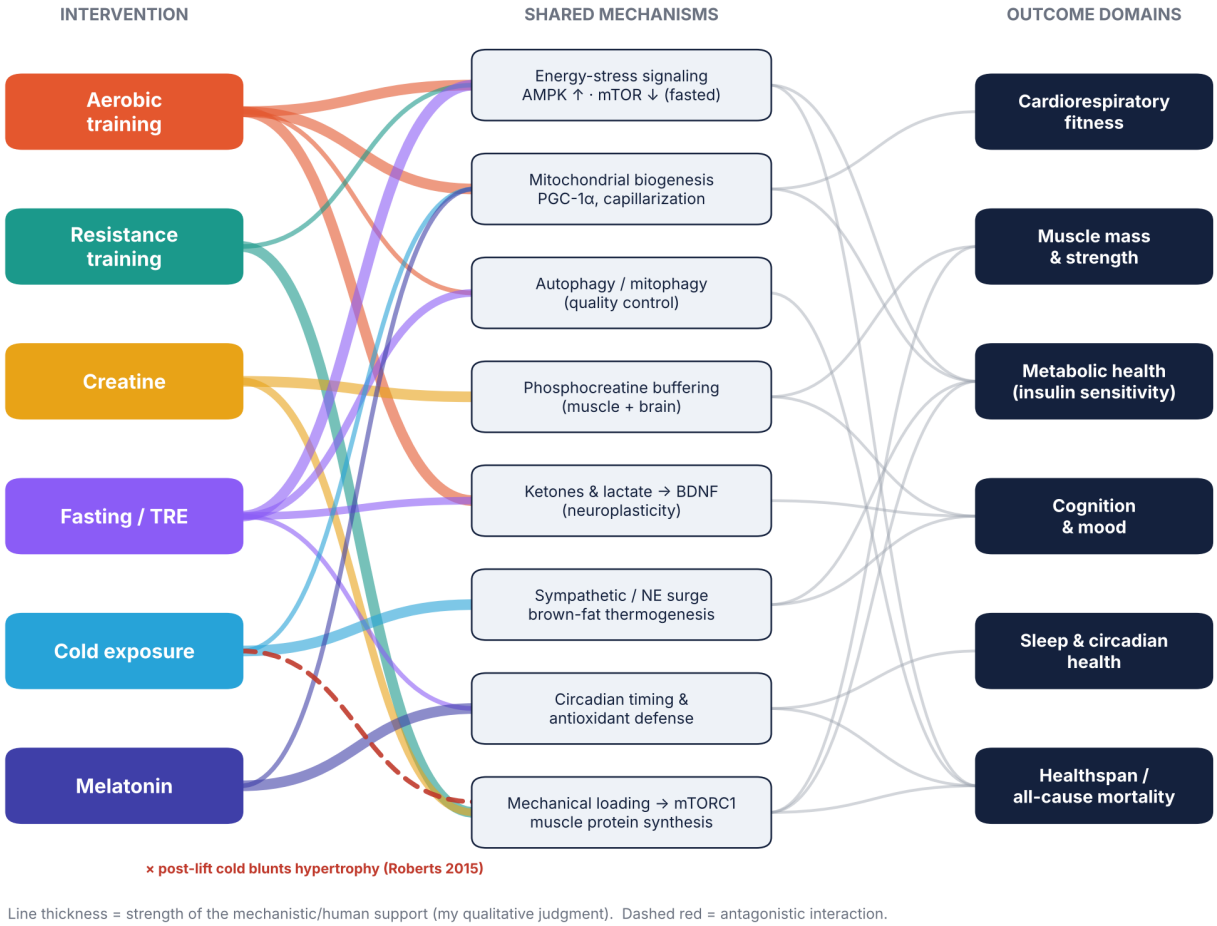


Figure 2. From interventions to shared mechanisms to outcomes. Where several colored lines land on the same mechanism box, the interventions share that pathway, which is where synergy (and redundancy) is most likely to arise.

Two readings of this map are useful. First, **convergence cuts both ways**: inputs to a saturating node (AMPK/PGC-1α is the obvious candidate) are likely to be sub-additive, so stacking several energy-stress signals will probably not multiply benefit. Second, the **one dashed line** (cold applied after lifting) is a reminder that some pairings are antagonistic at the level of a single pathway.

2 Aerobic training

Aerobic training

Builds the engine: stroke volume, capillaries, mitochondria. The strongest hard-outcome evidence in the stack.

Mechanism. Muscle contraction raises the AMP:ATP ratio and calcium flux, activating AMPK, p38 MAPK and CaMK. These converge on PGC-1α, the master regulator of mitochondrial biogenesis, and promote capillarization. Centrally, repeated high cardiac output remodels the heart (stroke volume), the component of VO_2max that declines most with age. In the brain, high-intensity exercise raises lactate, which signals through MCT transporters and SIRT1 to PGC-1α/FNDC5 and hippocampal *Bdnf* (rodent)[31]; prolonged exercise also raises β-hydroxybutyrate, which inhibits HDAC2/3 at the *Bdnf* promoter[17].

Human evidence.

- In 122,007 patients undergoing treadmill testing, all-cause mortality fell stepwise with fitness, with **no upper limit of benefit**: elite versus low fitness adjusted HR 0.20 (0.16–0.24), elite versus high 0.77 (0.63–0.95). Low fitness carried a larger risk than smoking, coronary disease or diabetes (whose own aHRs were 1.41, 1.29 and 1.40)[1].
- A meta-analysis of 33 studies (102,980 participants) found each +1 MET of fitness (about +1 km/h running speed) was associated with 13% lower all-cause mortality and 15% lower CHD/CVD events; the low-fitness group (below about 7.9 METs) had a 70% higher risk of death than the high-fitness group (about 10.9 METs or more)[2].
- Fitness is modifiable: in 579 men, each 1 mL/kg/min smaller decline (or larger gain) in VO_2max over 11 years was associated with 9% lower all-cause mortality[34].

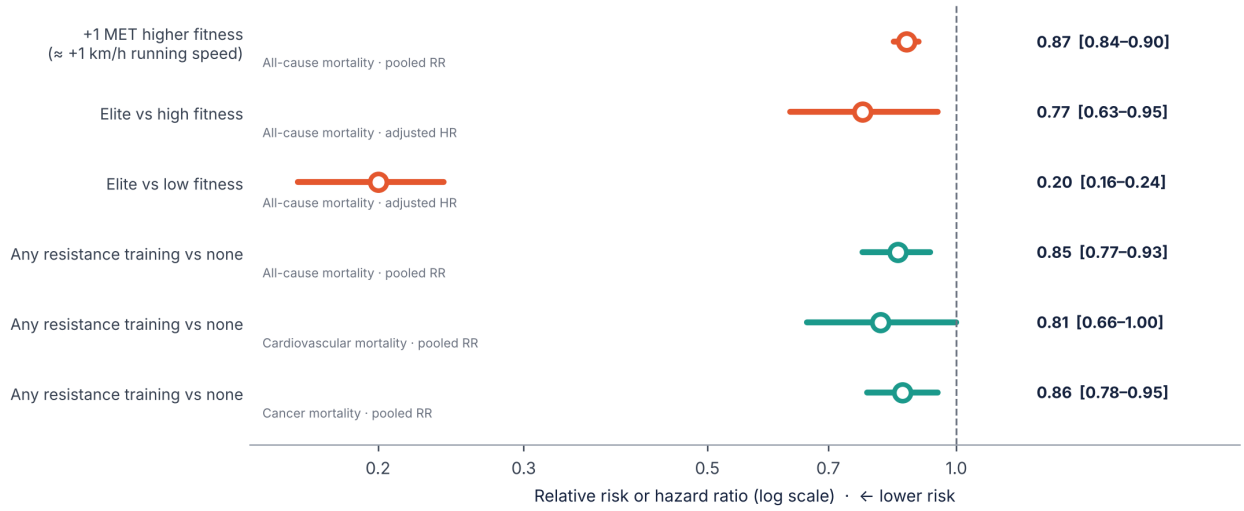


Figure 3. Mortality associations for aerobic fitness (orange) and resistance training (green). These are observational associations from different comparisons (per-unit, extreme-vs-reference, any-vs-none), so rows are not directly comparable. Confounding (healthier people are fitter) is the standing caveat.

Intensity versus volume. Matched for total work, PGC-1α induction was greater with higher intensity (Egan 2010[28]), and single-leg interval training beat continuous training for citrate synthase activity and mitochondrial respiration in the same subjects, though other mitochondrial markers rose equally (MacInnis 2017[29]). But the dose-response is an inverted U: after the highest-load week of a four-week progressive program, healthy volunteers showed impaired mitochondrial function and glucose tolerance, which mostly recovered after a lighter week (Flockhart 2021†[30]). Intensity is the more potent signal per minute, and the one that damages if overdosed; low-intensity volume is the way to add stimulus without paying that cost. This is the physiological logic of the polarized, roughly 80/20 pattern: most sessions at conversational pace (zone 2), one or two high-intensity sessions a week.

Practical parameters from the literature (synthesis)

3–4 zone-2 sessions of 45–90 min below the first lactate threshold; 1 session of VO_2max intervals (for example 4×4 min at 90–95% HRmax); hard cap of about 2 high-intensity sessions per week. Intervals are better done fed, because session quality is the point of them.

3 Resistance training

Resistance training

Protects the organ that actually fails with age: skeletal muscle. Small dose, large return.

Mechanism. Mechanical tension and stretch activate mTORC1 (through PI3K/Akt and mechanosensors), satellite cells and ribosome biogenesis; the result is muscle protein synthesis that, repeated, becomes hypertrophy. High-force work is the only stimulus that reliably recruits type II fibres, which are preferentially lost with ageing. Muscle is also the largest insulin-stimulated glucose sink, so preserving it protects insulin sensitivity[4].

Human evidence.

- A systematic review of 10 studies found any resistance training was associated with 15% lower all-cause mortality (RR 0.85, 0.77–0.93), 19% lower cardiovascular mortality (0.81, 0.66–1.00) and 14% lower cancer mortality (0.86, 0.78–0.95)[3].
- A dose-response meta-analysis of 16 cohort studies found 10–17% lower risk of all-cause mortality, CVD, total cancer, diabetes and lung cancer, *adjusted for aerobic activity*. The relationship was J-shaped, with the largest reduction at about 30–60 minutes a week; the effect of much higher volumes was unclear, and the certainty of evidence was graded very low for most outcomes[4].
- Combined muscle-strengthening and aerobic activity was associated with a lower risk of all-cause, CVD and cancer mortality than either alone[4].

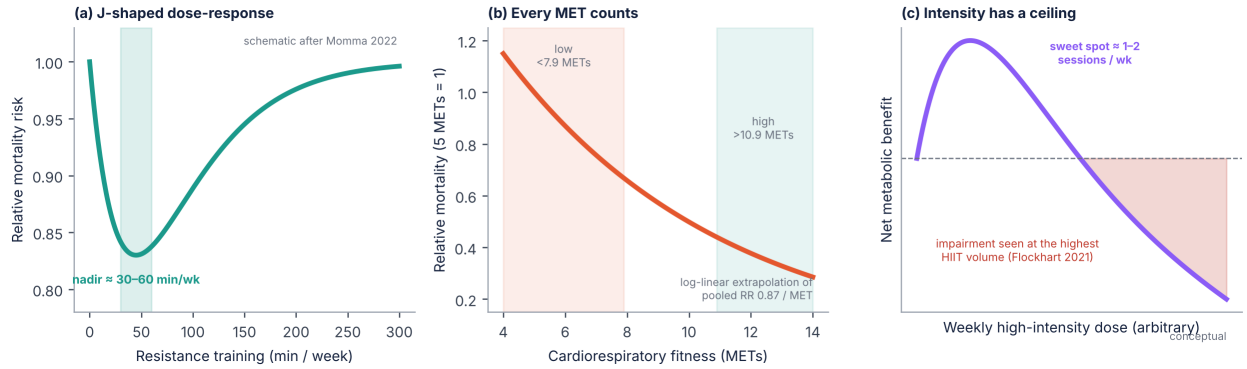


Figure 4. Dose-response shapes. (a) and (c) are schematic illustrations of published findings (Momma 2022; Flockhart 2021), not data plots. (b) extrapolates the pooled risk ratio per MET log-linearly; the true curve is uncertain at the extremes.

Caveats. All of this is observational; people who lift are different from people who do not, and the J-shape rests on few studies. Nevertheless, the dose is small enough, and the biological rationale strong enough, that the asymmetry favors doing it: two sessions a week covering all major muscle groups is the standard guideline.

4 Creatine

Creatine

An energy buffer for muscle and brain. Strong for strength and lean mass; modest, context-dependent for cognition.

Mechanism. Phosphocreatine (PCr) regenerates ATP from ADP in milliseconds via creatine kinase and acts as a spatial shuttle between mitochondria and ATP-consuming sites (Figure 5). The brain uses the same system and synthesizes its own creatine, but uptake across the blood–brain barrier is transporter-limited (SLC6A8), which caps how far oral supplementation can raise brain levels.

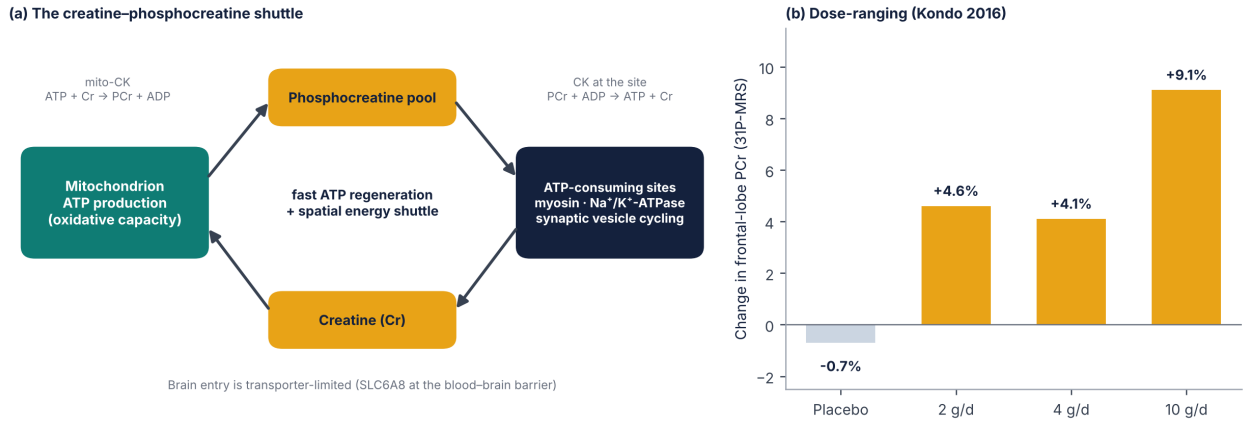


Figure 5. (a) The creatine–PCr shuttle. (b) Change in frontal-lobe PCr by 31P-MRS across doses in the only randomized dose-ranging study (Kondo 2016: 8-week placebo-controlled trial in females aged 13–20 with SSRI-resistant depression; the group differences were not statistically significant, $p = 0.69$).[9]

Human evidence.

- **Muscle and function.** A 2025 narrative review by Moriarty and colleagues concluded that creatine monohydrate, primarily combined with exercise training, is safe and improves lean mass, regional muscle size, strength, bone area and thickness, and functional ability in older adults; it also reports effects on glucose kinetics, cognition and memory[5].
- **Cognition, aging.** A systematic review of older adults found 5 of 6 studies reported a positive association with cognition, especially memory and attention, but four of the six were observational dietary-recall studies and the evidence quality was mostly low[6].
- **Cognition, mixed.** A Frontiers in Nutrition meta-analysis of 16 studies (492 participants) found benefit for memory, attention time and processing speed, with the attention effect significant at ages 18–60 but not above 60[7]. A 2026 commentary argues that some pooled effect sizes were double-counted[36], and in 2024 the European Food Safety Authority's panel concluded that a cause-and-effect relationship between creatine and improved cognitive function had not been established[37]. I treat the healthy-adult cognition literature as small effects of uncertain size.
- **Under stress.** The most consistent signal is energetic stress. A single high dose (0.35 g/kg) during a night of sleep deprivation protected PCr/Pi in motor regions of the brain and pH, and improved processing speed, in 15 healthy adults (Gordji-Nejad 2024)[8].
- **Brain dose.** Small studies suggest oral monohydrate raises brain total creatine by about 3–10%, with large measurement error, typically at 5–20 g/day (I could only read secondary summaries of those studies). In one small dose-ranging trial, mean frontal PCr rose 9.1% in the 10 g/day arm versus a 0.7% fall on placebo, though the differences between groups were not significant[9]. Most protocols for muscle use 3–5 g/day.

Caveats and open questions. Serum creatinine rises with supplementation without implying kidney injury (cystatin C is the cleaner marker if renal function matters). A 2026 study reported that creatine supplementation promoted metastasis in mice through a megakaryocyte–CKB–STAT5B pathway that produced hyperactive platelets, with increased platelet activity in healthy volunteers[33]. It is preclinical (experts have

criticized the tail-vein model and the dose), but worth tracking. In rodents, creatine reduced sleep need and homeostatic sleep pressure, while brain PCr only trended upward (Dworak 2017[32]); whether this matters in humans is unknown.

5 Fasting and time-restricted eating

Fasting and time-restricted eating

Metabolic switching: lower insulin, higher ketones, an AMPK-leaning state. The mechanism is richer than the human outcome data.

Mechanism. As feeding stops, insulin falls, liver glycogen is drawn down, and ketone production rises (Figure 6a). Reduced glucose and amino acids lower mTOR activity and raise the AMP:ATP ratio, activating AMPK and autophagy. β-Hydroxybutyrate is both a neuronal fuel and a signaling molecule that induces *Bdnf*. The review that framed this as "intermittent metabolic switching" highlighted induction of BDNF by β-hydroxybutyrate, mitochondrial biogenesis and autophagy as routes to neuroplasticity and stress resistance[16].

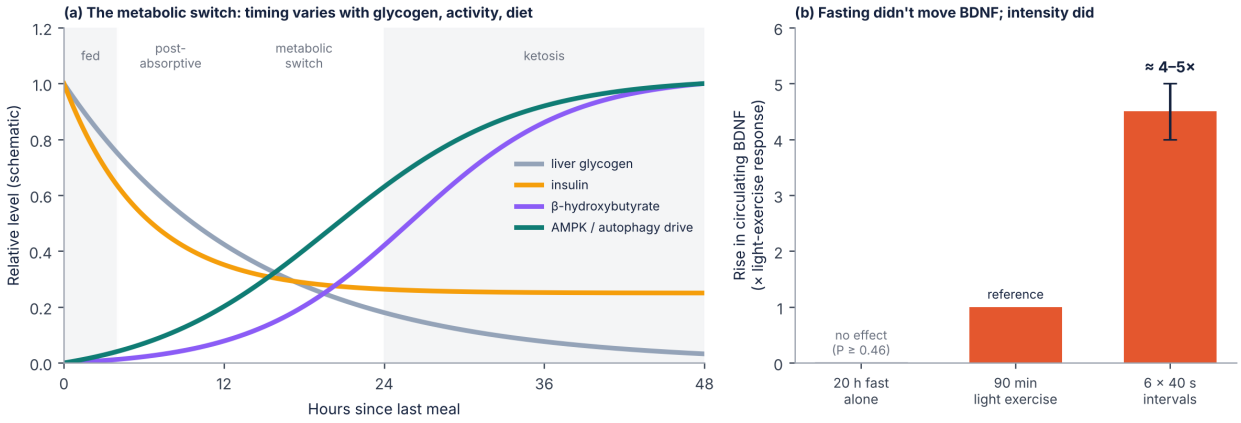


Figure 6. (a) Schematic of the fasting transition; timing varies by person. (b) In 12 volunteers, 20 h of fasting raised ketones (about 9-fold more ketone delivery to the brain) but did not change circulating BDNF; six 40-second intervals raised it 4–5-fold more than light exercise did[18].

Human evidence.

- In a 99-person trial of 8-hour eating windows (early, late, or self-selected) against usual care, the early and late groups kept significantly greater weight loss at 12 months, though only about a third reported still practicing TRE on their own at follow-up (a press-release figure)[19].
- When calories were equated, time-restricted eating was no more effective than calorie restriction alone for weight loss in a randomized trial (-8.0 versus -6.3 kg at 12 months, $p = 0.11$)[20], and a meta-analysis of 11 trials found intermittent and continuous energy restriction gave similar weight loss and cardiometabolic outcomes[21]. The metabolic gain appears to track overall energy reduction more than timing.
- Neuroprotection and longevity claims rest mainly on animal studies and short human trials; reviews are explicit that animal mechanisms are not assumed to translate[16].

Caveats. Lean-mass preservation requires adequate protein inside the feeding window and resistance training. Fasting is inappropriate with a history of disordered eating, and needs medical input with insulin or sulfonylurea use, pregnancy, or low body weight.

6 Cold exposure

Cold exposure

A sympathetic jolt and a small thermogenic adaptation. Good for alertness; antagonistic to lifting adaptations when mistimed.

Mechanism. Skin cold receptors drive sympathetic activation: noradrenaline rises sharply, vessels constrict, and shivering and non-shivering (brown-fat, UCP1-mediated) thermogenesis increase. Immersion at 14 °C for an hour raised metabolic rate 350%, plasma noradrenaline 530% and dopamine 250% without raising cortisol[10] (Figure 7a). The same group's earlier work found no significant dopamine rise, and these are peripheral plasma values, not brain dopamine, so brain-level claims go beyond the data.

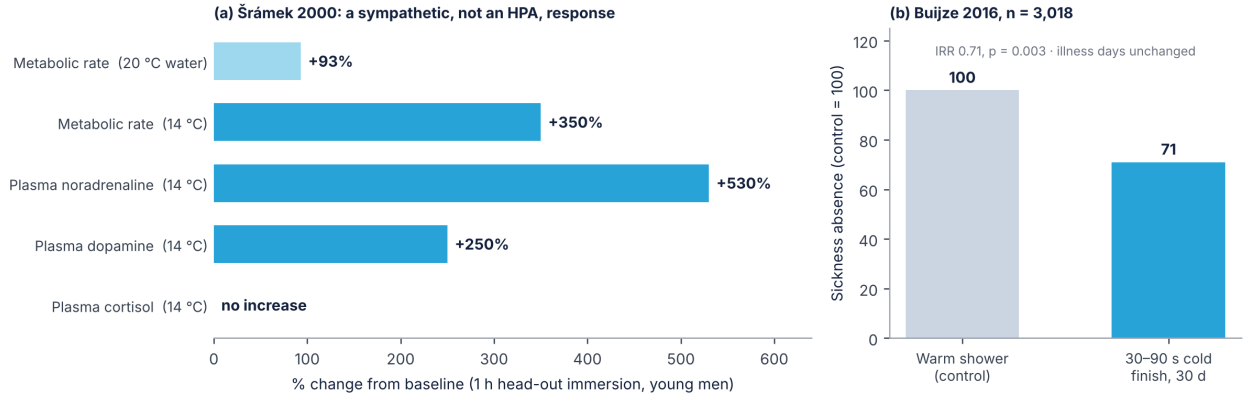


Figure 7. (a) Acute physiological response to 1 h head-out immersion[10]. (b) The only large randomized trial of cold showering: 30, 60 or 90 s of cold at the end of a daily shower for 30 days reduced sickness absence 29%; there was no dose-response and illness days did not change[11].

Human evidence.

- **Illness and energy.** Participants in the shower trial felt ill just as often but missed less work, which the authors attributed to the invigorating effect; a small quality-of-life gain at 30 days faded by 90[11].
- **Mood and stress.** A 2025 meta-analysis of 11 RCTs (n = 3,177) found stress was reduced only at 12 h after immersion (on few trials), null at other timepoints, and inflammatory markers were acutely higher[15].
- **Metabolism.** In people with active brown fat, 5–8 h of cold raised energy expenditure, glucose disposal and insulin sensitivity[13]; 10 days of cold acclimation improved peripheral insulin sensitivity in patients with type 2 diabetes[14]. The brown-fat contribution is small: measured thermogenesis is on the order of 10–25 kcal/day, rising two- to threefold with acclimation[38], so cold is not a weight-loss tool.
- **Habituation.** Across pooled studies (mean about 9 immersions, range 6–18) the heart-rate response to the initial shock fell by about 14 beats/min and respiratory frequency by about 8 breaths/min[35], which is the main safety adaptation.

The recovery–adaptation paradox. In a 12-week trial, 21 men who put their legs in 10 °C water for 10 minutes after each lifting session gained less muscle mass and strength than those who did active recovery; in a separate nine-man study, p70S6K phosphorylation and satellite-cell markers were lower after cold immersion[12]. Hot-water immersion did not show this penalty in the review literature. Cold is therefore best used away from lifting, or when speed of recovery matters more than growth.

Risks. The cold-shock response (gasp, hyperventilation, arrhythmia) is most dangerous in the first minutes and in cold open water; people with cardiovascular disease should be cautious. Showers carry far less risk than immersion.

7 Melatonin

Melatonin

A timing signal first, an antioxidant second, and a long-term safety question mark.

Mechanism. At low doses (about 0.5 mg in phase-response studies[40]) melatonin acts as a chronobiotic through MT1 and MT2 receptors, shifting the phase of the circadian clock according to when it is taken; it is a timing cue rather than a sedative. At higher doses it is reported to accumulate in mitochondria, scavenge reactive oxygen and nitrogen species, and support electron-transport function[26], but that work is mostly preclinical and a review found the evidence insufficient to recommend an effective dose[39]. The two ranges are different pharmacologies (Figure 8b).

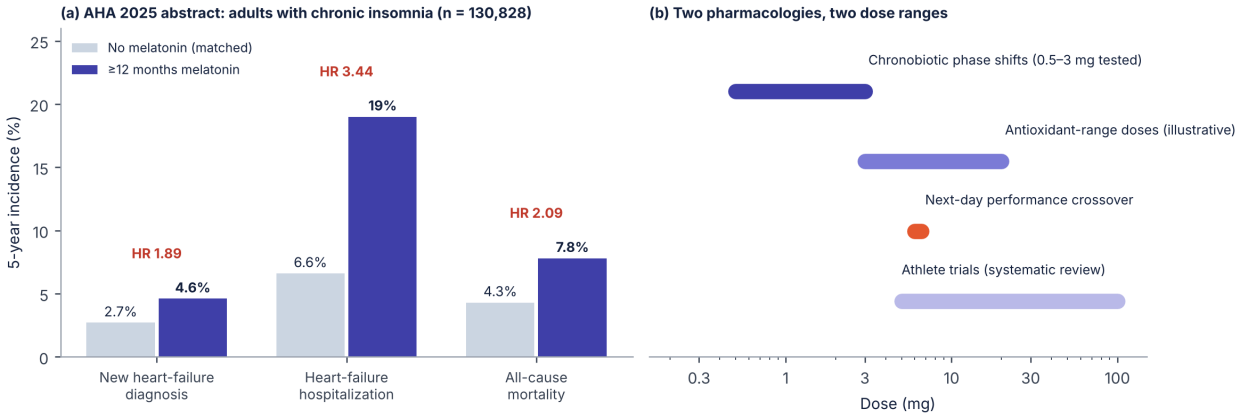


Figure 8. (a) Observational data from the AHA 2025 abstract (TriNetX electronic records, 65,414 users matched to 65,414 non-users)[22]. (b) Dose ranges used in different literatures; the antioxidant range is illustrative, since no effective human dose has been established[39].

Human evidence.

- **Exercise performance.** A meta-analysis of 19 RCTs (266 participants) found evening melatonin, with a gap of more than 6 hours before exercise, improved endurance and explosive power and reduced creatine kinase[23]. In one crossover study of 12 trained men, 6 mg at night improved next-day high-intensity performance and perceived recovery for up to 72 h[24]. A separate systematic review of highly trained athletes (doses 5–100 mg) found mixed, inconsistent effects on strength and power and no clear benefit for speed[25].
- **Circadian effects of exercise.** One hour of evening exercise advanced circadian phase by about 30 minutes versus a delay of 25 minutes without exercise in young men, though the shift was attenuated on the second day[27], so training time and melatonin time may interact.
- **Safety signal.** In adults with insomnia taking melatonin for 12 months or longer, 5-year incidence of heart failure was 4.6% versus 2.7% (HR 1.89), heart-failure hospitalization 19% versus 6.6% (HR 3.44), and all-cause mortality 7.8% versus 4.3% (HR 2.09)[22]. This is a conference abstract that has not been peer reviewed. The authors stressed the study cannot show causation; likely confounders include insomnia severity and over-the-counter use of melatonin among controls, which electronic records cannot capture. An industry group's response noted decades of experience with low-dose, short-term use. This is a signal to track, not a verdict.

Practical implication. Low-dose, well-timed use matches the mechanism and the benign short-term record. Indefinite high-dose use in the evening is the pattern with the least supporting evidence and the one the new observational signal concerns.

8 Pairwise interactions

Fifteen pairs, color-coded. Only two earn the label "synergy" on current evidence (aerobic with resistance, resistance with creatine); one is a clear conflict (cold after lifting). Most of the remainder are timing problems, which are solvable.



Figure 9. Interaction matrix (my synthesis). Symbols repeat the colors for black-and-white printing.

Resistance + creatine	Best-evidenced pairing: lean mass and strength gains exceed training alone, especially in older adults[5].
Aerobic + resistance	Concurrent training is associated with lower mortality than either alone[4]; interference is real but modest, and sequencing manages it.
Resistance + cold	Post-lift cold blunts hypertrophy and anabolic signaling[12]. Separate by several hours or skip.
Resistance + melatonin	Antioxidant doses near training may blunt the oxidative signals that drive adaptation (theoretical); keep melatonin away from training.
Fasting + melatonin	Circadian-phase effects on glucose handling are proposed[22]; avoid taking melatonin close to meals.
Fasting + cold	Plausibly additive metabolic stress, with hypoglycemia, dizziness and shivering risks; untested.
Creatine + fasting	Calorie-free, so timing is flexible; taking it with the first meal is a convenient default.

9 Hypotheses: what might they do together?

A caution before the speculation

Hazard ratios do not multiply. The fitness, strength, fasting and cold effects above come from different populations, outcomes and study types; stacking them arithmetically ("$0.85 \times 0.87 \times \ldots$") would double-count shared pathways and confounding. Treat every hypothesis below as a testable guess, not a forecast. Ratings: left dots = mechanistic plausibility, right dots = human evidence for the *combination*.

H1. Foundation plus modulators

PLAUSIBILITY ●●●●● COMBINATION EVIDENCE ●●●○○

Aerobic fitness and muscle set the *ceilings* (VO_2max, lean mass, strength) that dominate mortality data. Creatine, fasting, cold and melatonin act on the *approach to the ceiling*: training volume and recovery, metabolic flexibility, arousal, timing. **Prediction:** a person who adds all four modulators but under-trains the foundation will do worse than one who trains well and adds none. This is the hypothesis best supported by the effect-size comparison in this report.

H2. Pulsatile signaling: alternate catabolic and anabolic states

PLAUSIBILITY ●●●●○ COMBINATION EVIDENCE ●●○○○

Fasting and zone-2 work push the cell toward AMPK, autophagy and mitochondrial turnover; feeding plus lifting (with creatine supporting training capacity) pushes toward mTORC1 and tissue building. Cycling the two within a day may give more of both than a constant fed state gives of either (Figure 10). **Prediction:** on a time-restricted schedule with adequate protein and lifting, lean mass is preserved while insulin sensitivity and markers of mitochondrial quality improve. **Against:** time-restricted eating alone does not beat calorie restriction when calories are matched[20].

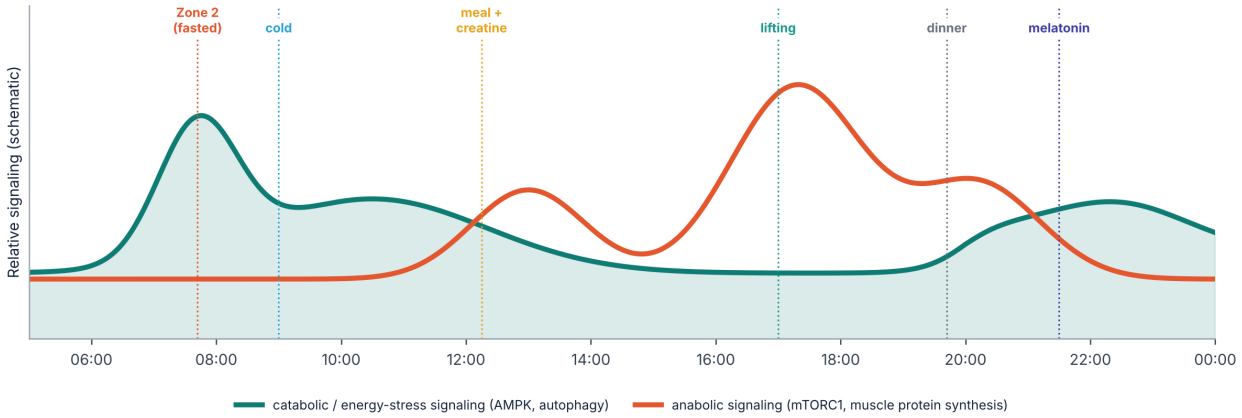


Figure 10. Conceptual signaling profile across a day (not measured data). Fasted low-intensity work lengthens the catabolic phase; the feeding window and lifting create the anabolic pulse.

H3. Brain plasticity convergence

PLAUSIBILITY ●●●○○ COMBINATION EVIDENCE ●○○○○

Four inputs reach the same neural machinery by different routes: exercise intensity (lactate → BDNF), prolonged exercise or fasting (β-hydroxybutyrate → HDAC inhibition at *Bdnf*)[17,31], noradrenergic arousal from cold, and a larger energetic reserve from creatine under stress[8], with melatonin-protected sleep to consolidate. **Prediction:** learning scheduled shortly after intense exercise and followed by good sleep improves more than learning alone. **Against:** 20 h of fasting alone did not change circulating BDNF[18]; creatine effects in healthy adults are small[7]; cold's attention effects are mixed. Serum BDNF is also a poor proxy for brain BDNF. The convergence may mostly mean redundancy rather than addition.

H4. Thermo-metabolic additivity

PLAUSIBILITY ●●●○○ COMBINATION EVIDENCE ●○○○○

Cold acclimation recruits brown fat and improves insulin sensitivity modestly[13,14]; time-restricted eating improves glycemic control; aerobic training raises mitochondrial capacity. **Prediction:** the combination improves glucose disposal more than any one alone, with cold contributing the smallest share given brown fat's small energetic footprint[38]. **Test:** OGTT or clamp after 8–12 weeks.

H5. A circadian scaffold

PLAUSIBILITY ●●●●○ COMBINATION EVIDENCE ●●○○○

Three of the interventions are zeitgebers: timed exercise shifts the clock[27], restricted feeding anchors peripheral clocks, and low-dose melatonin shifts phase by its timing. Aligned, they may reinforce each other; scattered, they may fight (late hard exercise delays; late cold arouses; late meals plus melatonin impair glucose handling). **The missing zeitgeber is morning light**, which is the strongest of all and is not on the list. **Prediction:** a consistent daily sequence beats the same interventions at random times.

H6. The hormesis budget

PLAUSIBILITY ●●●●○ COMBINATION EVIDENCE ●●○○○

Cold, fasting, hard intervals and antioxidant dosing are all stressors that are beneficial in a window and harmful beyond it. The inverted U seen for HIIT volume[30] may apply to the *total* stress load. **Prediction:** resting heart rate, heart-rate variability and sleep quality drift before performance does, so they work as early warnings. **Rule of thumb:** place different stressors on different days or hours rather than stacking them.

10 A worked example: one possible day

The chart applies the interaction rules from Chapters 8–9. It is a template to adapt, not a prescription. Resistance and aerobic sessions are placed on separate occasions; cold is kept away from lifting; creatine rides with the first meal; melatonin is taken low-dose, well after the last meal and training.



Figure 11. Hypothetical lift-day template (16:8 eating window). On interval days the zone-2 slot would move to the afternoon, fed.

Measuring whether it works

Each hypothesis predicts something measurable. A small dashboard turns the stack into an experiment:

Domain	Measure	Why
Aerobic capacity	VO_2max or 12-min test; resting HR	Strongest mortality gradient; every MET counts
Muscle	Grip strength; DEXA lean mass; a fixed lift	Function, not appearance, predicts outcomes
Metabolic	Fasting glucose, insulin, HbA1c; ApoB	Tests H2 and H4
Recovery / load	HRV, sleep duration and regularity	Early warning for H6; tests H5
Brain	A repeatable cognitive test at a fixed time	Tests H3 (noisy; use trends)
Safety	Blood pressure, cystatin C if on creatine, symptoms	Catches the failure modes above

11 Limits of this report

- **Observational dominance.** Most hard-outcome data (fitness, resistance training, melatonin) are cohort studies; healthy-user bias and reverse causation apply in both directions.
- **Rodent dominance in the mechanisms.** Brain BDNF routes, creatine's signaling effects, and melatonin's mitochondrial actions are mostly animal or in vitro findings.
- **Combinations are untested.** Chapter 9 is reasoning, not evidence. The correct summary of the combined benefit is: plausible, probably partly redundant, probably more sensitive to timing than to inclusion.
- **Scores and diagrams are qualitative.** Scorecard grades, line thicknesses and schematic curves are my judgment, and where primary full texts were unreachable I relied on abstracts, repository records, and for two items (the AHA abstract and a 12-month eating-window follow-up) conference and press coverage.

- **Medical context.** Individual risks (cardiovascular disease, medications, history of eating disorders, mood disorders, kidney disease) change the calculus; a clinician who knows the patient should have the final say.

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† Not checked against the primary full text: the Flockhart design details were read only through the abstract and a secondary summary, the 12-month eating-window follow-up and the AHA abstract through press and conference coverage, and the Burgess citation from a search listing. For the other entries I read the paper's abstract or journal or repository page; verify exact figures against the full papers before citing.